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Python Code Debugging Using PyCharm
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**** Debugging is the process of finding, understanding, and fixing errors in a program.**

**** When we write a Python program, it may not always produce the expected result. The program may:**

- Stop suddenly
- Display an error message
- Produce an incorrect output
- Execute the wrong block of code
- Get stuck in an infinite loop

**** Debugging helps us understand:**

- which line is executing
- what values variables contain
- why a condition is true or false
- where the error occurred
- how the program reached that point

Debugging means executing a program step by step to identify and fix problems.

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Why Should We Use a Debugger
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**** Using a debugger, we can:**

- pause program execution
- execute one line at a time
- inspect variable values
- enter inside functions
- skip function execution
- evaluate expression
- identify logical errors
- understand program execution flow

Normally, we execute the program using the Run option.

To debug it, we use the Debug option.

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What is a Breakpoint?
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**** A breakpoint tells PyCharm:**

- Pause the program before executing this line.

**** when the debugger stopped at break point, we can control program execution using below debugging shortcuts..**

Shift + F9 : Start Debugging

F7 : step into (enter inside function)

F8 : step over (move to next line)

F9 : Move to next break point in the flow (Resume program)

Shift + F8 : Step Out (Complete function and come out and return to the called line)

Alt + F9 : Run to cursor (move the debugger to the line where our cursor is pointed)

Ctrl + F2 : Stop Debugging

** When program in under debugging mode, we can change variable values in the runtime for testing some special scenarios...